

GREG DICRISTOFARO

Software Engineer

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SUMMARY

Highly accomplished **Full-Stack Software Engineer** with 8+ years of experience in enterprise software development. Expertise includes Java (Spring Boot), C++, Python, SQL, C# (ASP.NET Core) and modern front-end frameworks (React, Angular). Proven ability to design, test, and deploy high-performance, resilient applications leveraging cloud services (GCP, AWS) and modern DevOps practices (Docker, Jenkins)

EXPERIENCE

Software Engineer

Sleuth Kit Labs

📅 February 2020 - Present 📍 Somerville, MA

- Develop **Digital Forensics and Incident Response software**: Cyber Triage, Autopsy, and The Sleuth Kit
- Expanded and optimized **data pipeline processing** handling complex SQL queries over large data sets using Java, C++, Python, and cloud updatable rules over GCP
- **Modernized testing infrastructure** using Jenkins, Nginx, Docker, Linux, Bash, Python, Apache Ant, JUnit, and Mockito
- **Standardized multi-platform deployment (Windows, Linux, macOS)** through new Snap, Homebrew, and Advanced Installer packages, streamlining the installation process for end-users
- Provide internal and **customer documentation** for complex features
- Architected and built **RESTful and Reactive web applications** using Spring Boot, Swagger, Angular, and React (Material UI), leveraging Project Reactor and Server Sent Events (SSE) for real-time data streaming
- Developed **serverless and containerized cloud infrastructure** using Docker, JMS, ActiveMQ, AWS DynamoDB, AWS Lambda, GCP Cloud Functions, GCP BigQuery, and GCP Datastore
- Develop **Java Desktop Applications** with Swing, JavaFX, Maven, Apache Ivy, Apache POI, SQLite, PostgreSQL, and Netbeans RCP; **Contributed to ControlsFX**, a popular JavaFX controls library

SKILLS

Languages: Java, C, C++, Python, TypeScript, C#, Rust, Bash, SQL

Web/Backend: Spring Boot, React, Angular, ASP.NET Core, Project Reactor, Server Sent Events (SSE), WebAssembly

Cloud/DevOps: GCP (BigQuery, Cloud Functions, Datastore), AWS (Lambda, DynamoDB), Docker, Jenkins, Nginx, Linux

Databases: PostgreSQL, SQLite, MS SQL Server

Desktop: JavaFX, Swing

ACHIEVEMENTS

✓ Presenter

Presenter for OSDFCon 2021 (Autopsy Scoring: Finding the Relevant Data with Analysis Results)

✓ KBR Certificate of Appreciation

In recognition of outstanding performance & commitment to excellence (2019)

PROJECTS

constantqjs

Implemented the Constant-Q transform for spectral analysis using **C++ compiled to WebAssembly** (via Emscripten), integrating with **TypeScript/Angular** for real-time visualization (Chart.js, Web Audio API)
<https://github.com/gdicristofaro/constantqjs>

Software Engineer

KBR

📅 July 2017 - February 2020 📍 Lexington Park, MD

- Modernized large J2EE web applications built with JSP and JSF
- Create **data visualizations** with JavaScript, TypeScript, **D3.js**, and SVG to present complex data to customers
- Developed and maintained **Web-based UI and RESTful services** development with **React and ASP.NET Core**
- Queried large datasets using Microsoft SQL Server, T-SQL, Entity Framework, and Linq to SQL

Music Teacher

St. Mary's County Public Schools

📅 August 2006 - June 2017 📍 Leonardtown, MD

- Demonstrated strong organizational, leadership, and logistics skills by coordinating school-wide schedules and leading public-facing events

EDUCATION

Master of Science in Software Development

Boston University

📅 9/2017 - 5/2019 📍 Boston, MA 🎓 GPA 3.96 / 4.00

Software Engineering Certificate

Harvard University

📅 9/2012 - 12/2014 📍 Cambridge, MA 🎓 GPA 4.00 / 4.00

Master of Music in Music Education

University of the Arts

📅 6/2012 - 12/2015 📍 Philadelphia, PA 🎓 GPA 4.00 / 4.00

Thesis: Designed cross-platform music composition software to foster creativity for young students that processes audio, visualizes music in a sequencer and renders standard music notation

sci-fi-crawl

Sci-fi customizable opening credit text using React, TypeScript, Material UI, and CSS animations <https://github.com/gdicristofaro/sci-fi-crawl>

CalendarMaker

Calendar making application developed with TypeScript, React, Moment.js, Material-UI, and PptxGenJS
<https://github.com/gdicristofaro/CalendarMaker>

PSDtoSVG

Python web application for converting Photoshop files into SVG's for data visualization
<https://github.com/gdicristofaro/PSDtoSVG>

SeamCarve

Scala implementation of the Seam-Carving Algorithm for content-aware image resizing
<https://github.com/gdicristofaro/SeamCarve>

CSPSolver

Class scheduling program based on the AC-3 algorithm utilizing Apache POI API to import from Excel documents written in Scala
<https://github.com/gdicristofaro/CSPSolver>

SpellChecker

Spell-checking algorithm built in C checking words against a dictionary file utilizing a self-developed hash map with buckets
<https://github.com/gdicristofaro/SpellChecker>